

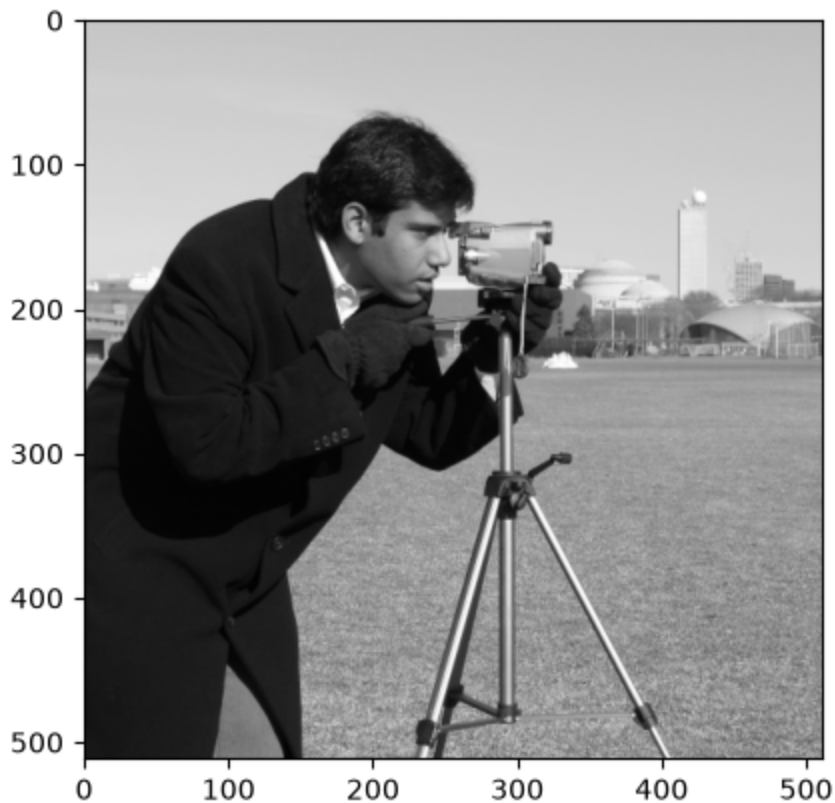
LAB 1

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Task 2: Loading and Visualizing Images

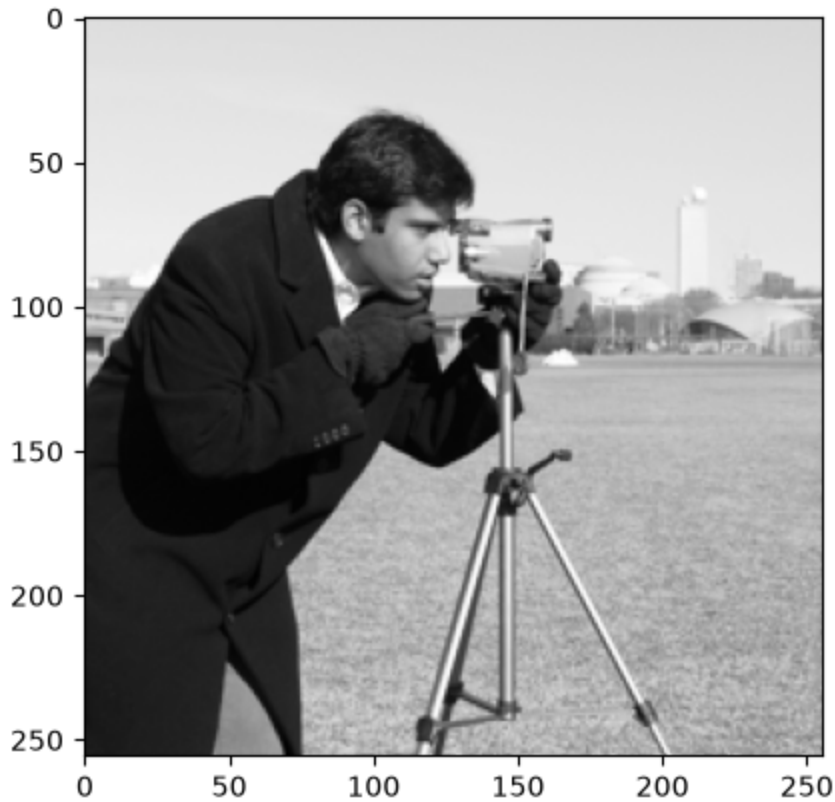
Loading/reading an image using OpenCV

```
In [1]: import cv2
img = cv2.imread('images/cameraman.tif') #Load the image data into a NumPy array
import matplotlib.pyplot as plt
plt.imshow(img)
plt.show()
```



Loading/reading image using PIL

```
In [2]: from PIL import Image
img2 = Image.open('images/lena_gray_256.tif')
import matplotlib.pyplot as plt
import matplotlib.cm as cm
plt.imshow(img2, cmap = cm.Greys_r)
plt.show()
```



Task 3: Image Storing

```
In [3]: # Saving image to disk using OpenCV
cv2.imwrite('NewImg.jpg', img)

# Saving an image to disk using PIL
img2.save('NewImg2.jpg')
```

Task 4: Display image as an Array

```
In [6]: print(img.shape) #Note: img loaded using open cv is a numpy array
print(img)

import numpy as np
# Convert the image to a NumPy array
img_array = np.array(img2)
# Print the array
print(img_array.shape)
print(img_array)
```

```

(512, 512, 3)
[[[200 200 200]
  [200 200 200]
  [200 200 200]
  ...
  [189 189 189]
  [190 190 190]
  [190 190 190]]]

[[[200 200 200]
  [199 199 199]
  [199 199 199]
  ...
  [190 190 190]
  [190 190 190]
  [190 190 190]]]

[[[199 199 199]
  [199 199 199]
  [199 199 199]
  ...
  [190 190 190]
  [190 190 190]
  [190 190 190]]]

...

[[ [ 25  25  25]
  [ 25  25  25]
  [ 27  27  27]
  ...
  [139 139 139]
  [122 122 122]
  [147 147 147]]]

[[ [ 25  25  25]
  [ 25  25  25]
  [ 26  26  26]
  ...
  [158 158 158]
  [141 141 141]
  [168 168 168]]]

[[ [ 25  25  25]
  [ 25  25  25]
  [ 27  27  27]
  ...
  [151 151 151]
  [152 152 152]
  [149 149 149]]]]
(256, 256)
[[[199 199 200 ... 190 190 190]
  [199 199 200 ... 190 190 190]
  [200 200 199 ... 191 191 191]
  ...
  [ 25  25  24 ... 136 143 118]]]

```

```
[ 26 25 26 ... 147 154 144]  
[ 25 25 25 ... 146 146 151]]
```